

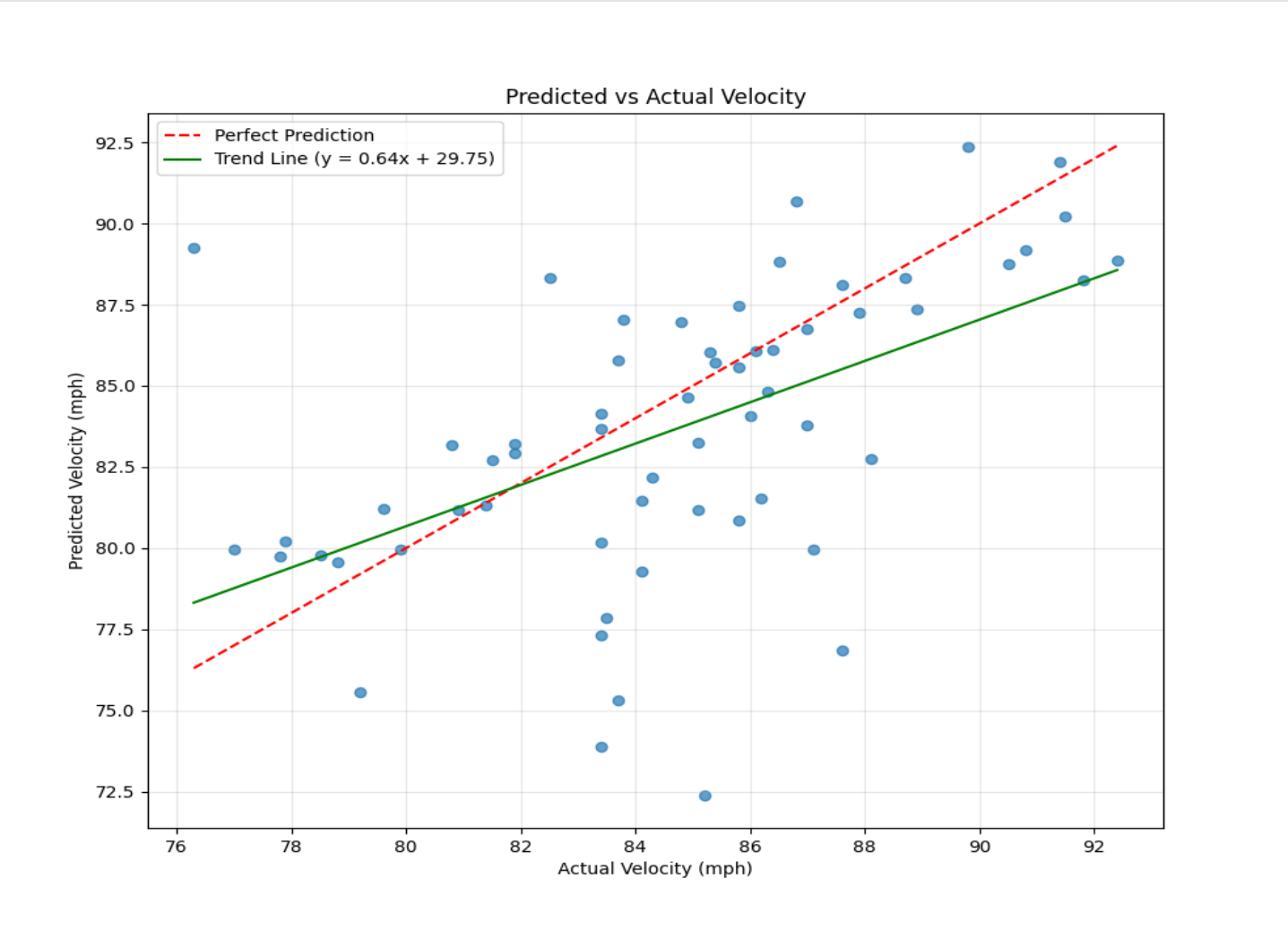
Transforming Pitching: A Data-Driven Approach to Mechanics and Coaching Using Deep Learning

Caleb Guisewite
Belmont University, Nashville Tennessee

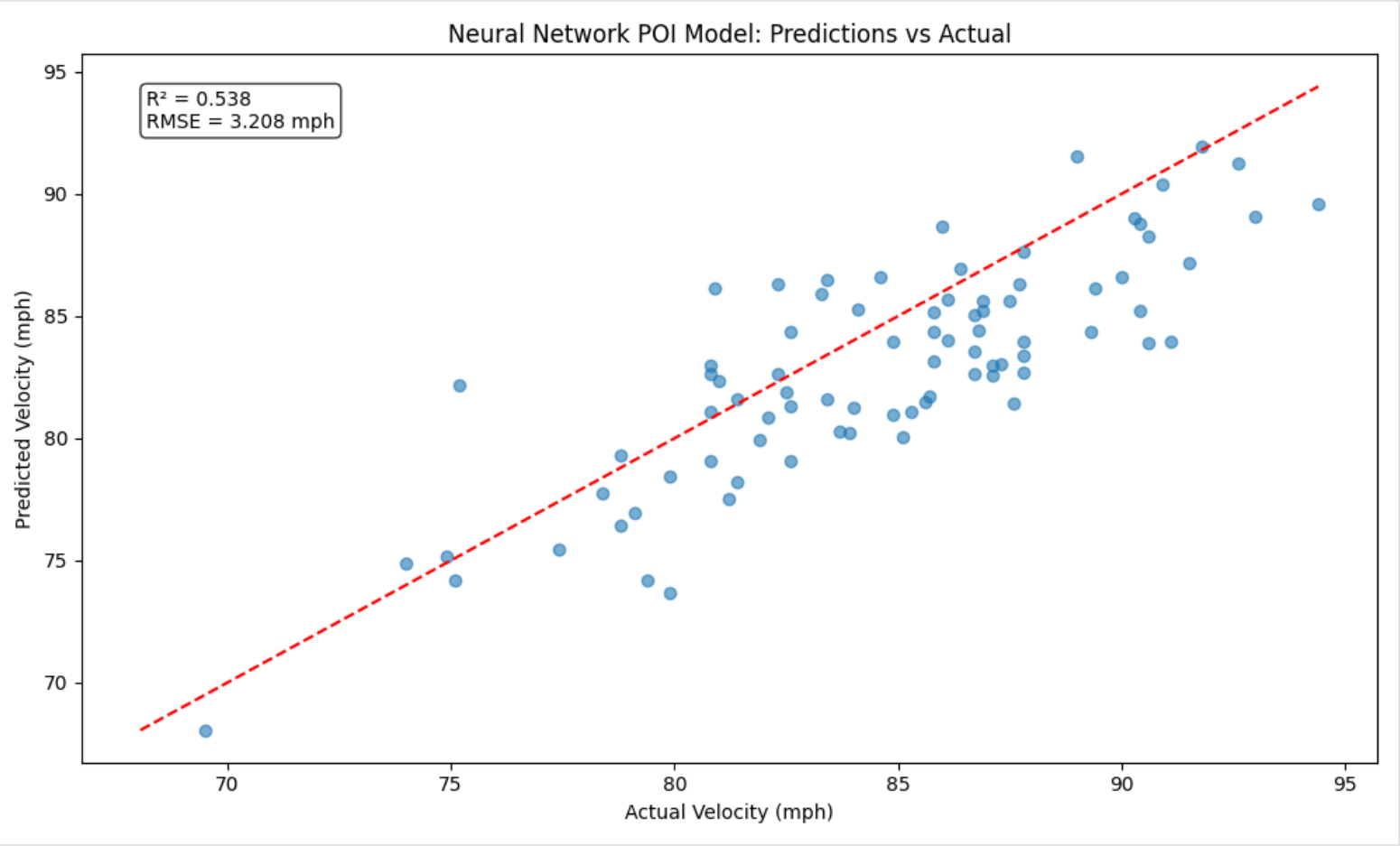
Introduction

Pitching is one of the most biomechanically demanding movements in sports, where small variations in mechanics can have major impacts on velocity, performance, and injury risk. This project explores the ability of deep learning models — particularly transformer-based architectures — to learn the complex, time-dependent relationships that underlie pitch velocity. By comparing traditional machine learning models using static data with advanced deep learning models trained on sequential motion capture data, we aim to uncover not only how fast a pitcher throws, but why — providing interpretable insights that could inform training, injury prevention, and player development.

Transformer Performance



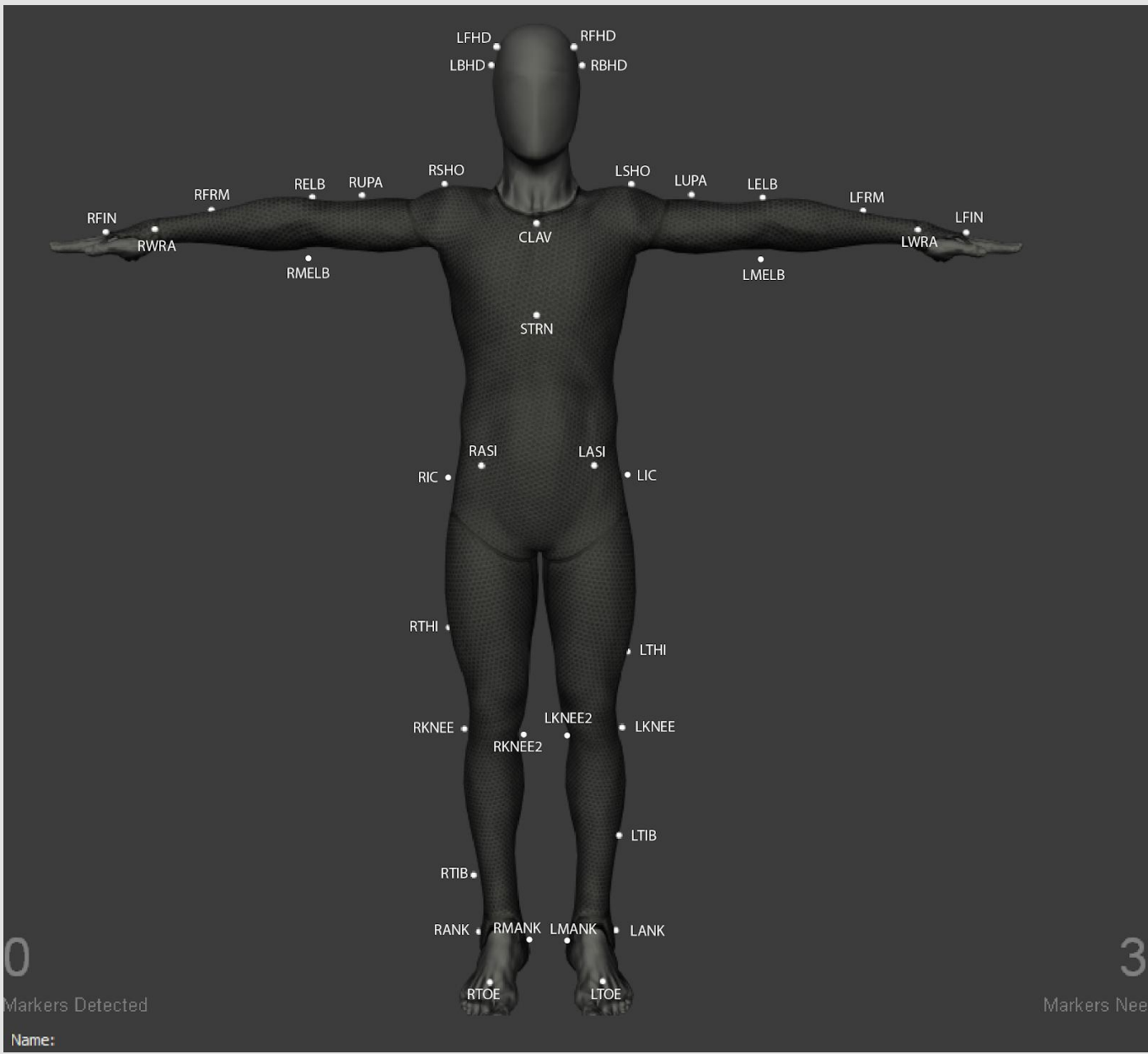
Comparing Models



Model Architecture

Insert Architecture picture here.

Data and Feature Overview

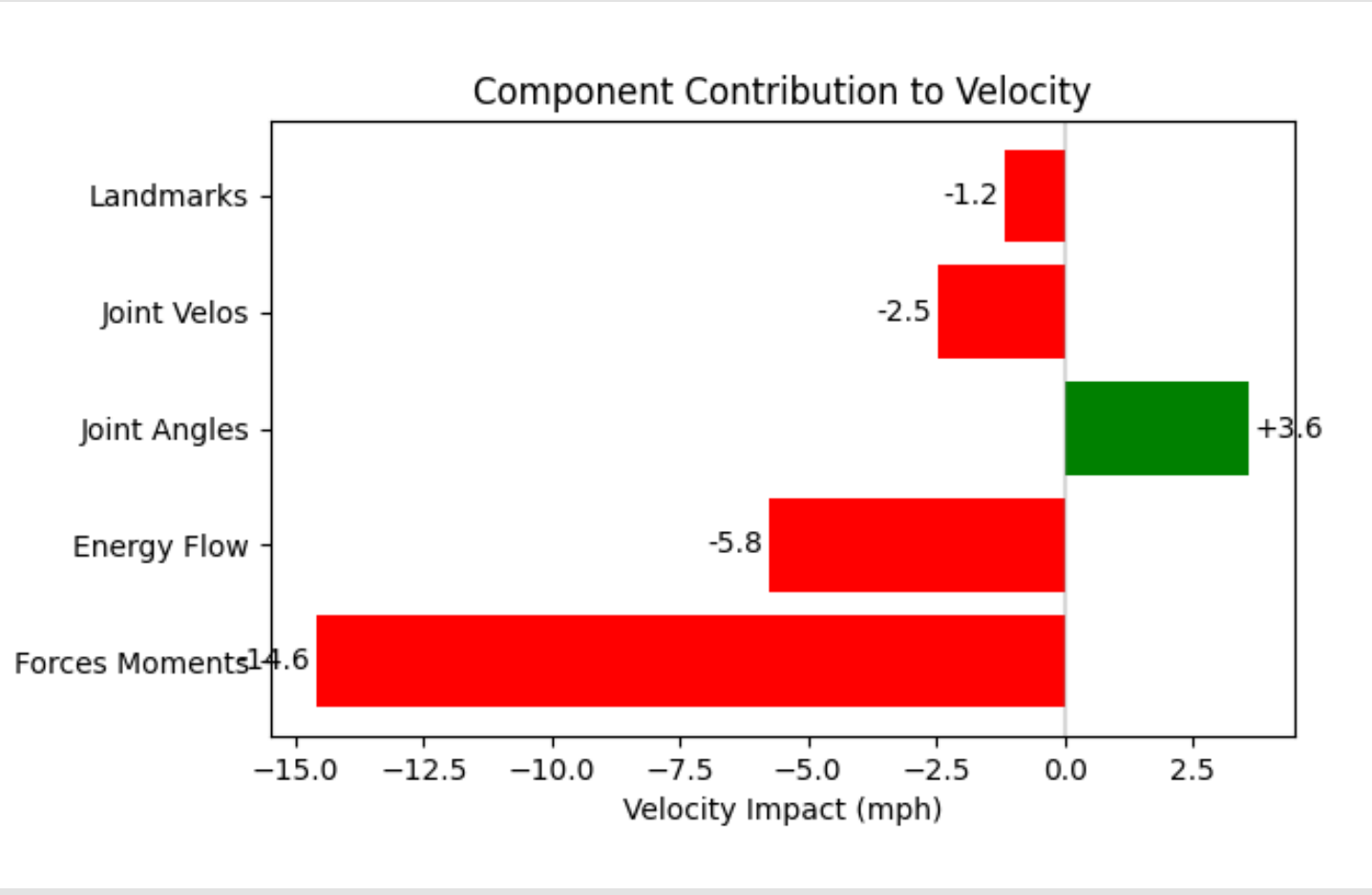
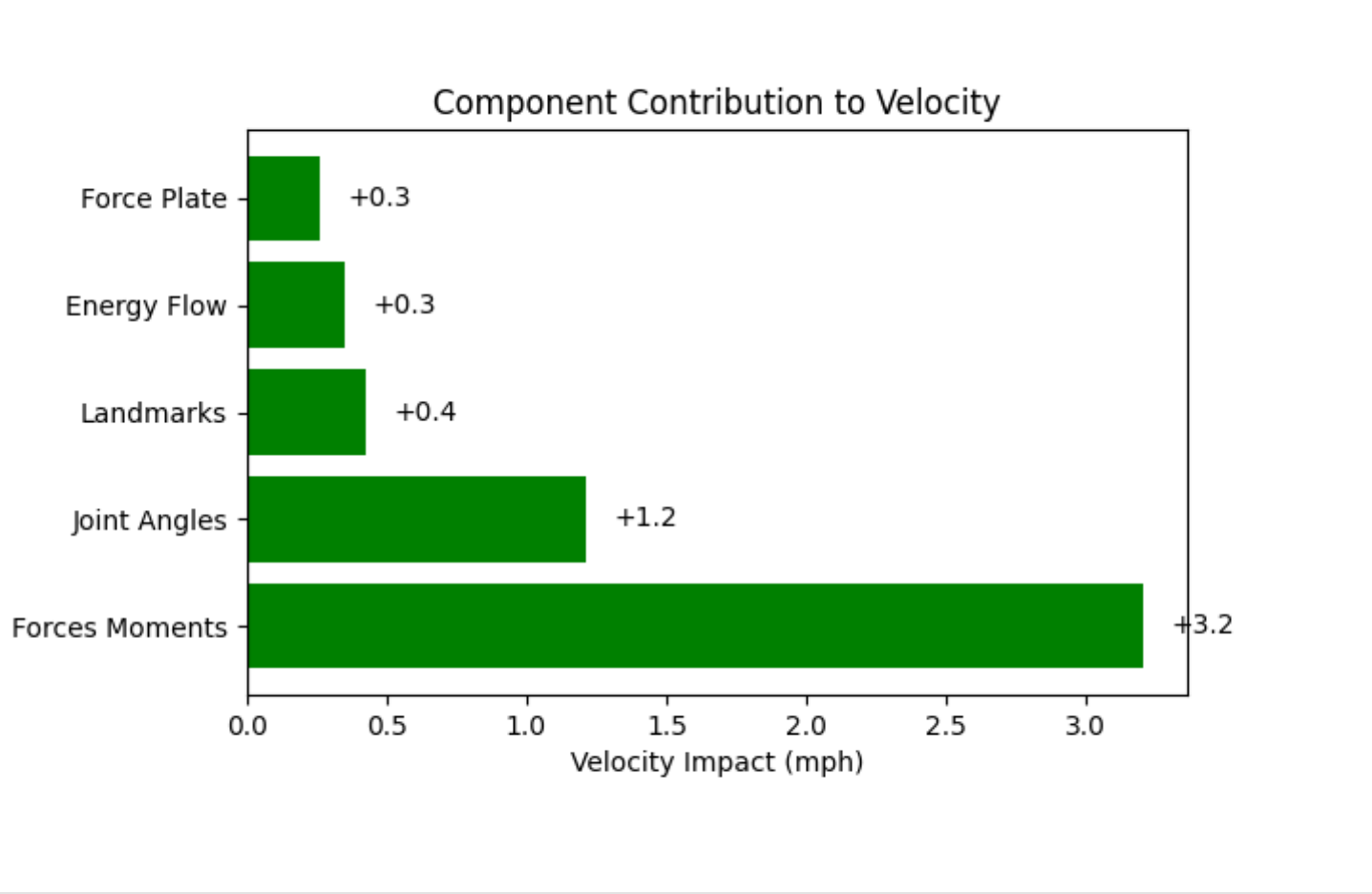


- The full sequential biomechanical data is divided up into 6 groups
- Energy Flow: energy transfer, generation, and absorption
 - Force Plate: Rear leg, lead leg ground reaction forces
 - Force Moments: Joint Forces and Moments
 - Joint Angles: Joint angles
 - Joint Velos: Joint angular velocities
 - Landmarks: Joint Center Positions

Next, we have static data

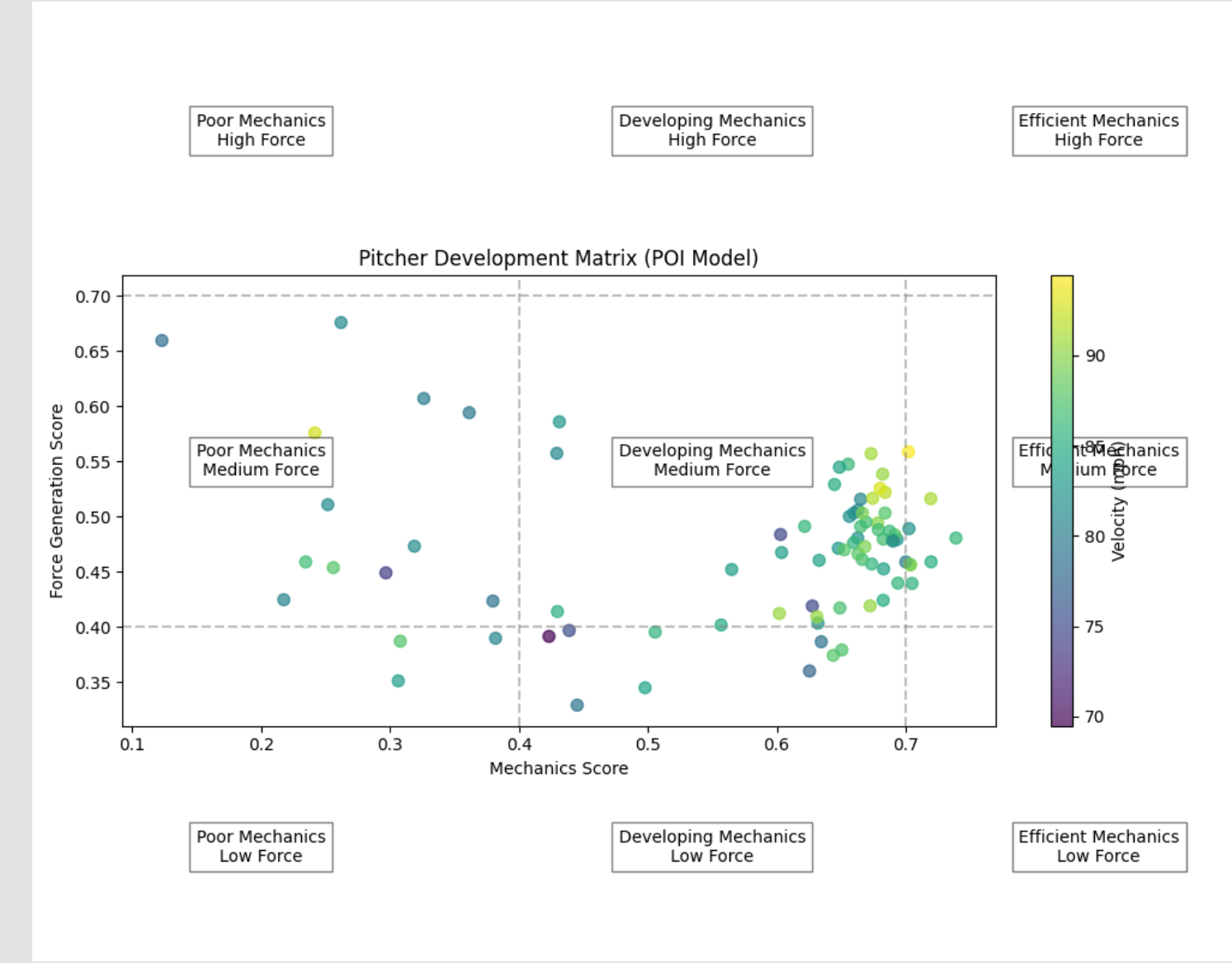
- Point of Interest Data: Kinematic, kinetic, and energetic metrics commonly referenced in biomechanical analyses

Coachable Insights



Dual Pathway Setup

Insert feature routing picture here.



Future of Baseball Analytics

As data collection in baseball evolves from stat sheets to high-resolution biomechanics, the next frontier in analytics lies not just in what happened, but how and why. Motion capture systems, force plates, and wearable tech now provide detailed measurements of an athlete's movement, enabling teams to analyze the mechanics behind every pitch, swing, and sprint. Incorporating biomechanics into predictive models opens the door to deeper, more personalized insights. Rather than relying solely on historical outcomes, teams can now project performance based on how a player moves — offering early signals of potential, inefficiency, or injury risk. Deep learning models, especially transformers, are uniquely positioned to harness these complex time-series inputs. They can learn hidden dependencies across joints, timing, and force application that traditional models overlook. As interpretability techniques improve, these models won't just predict velocity — they'll explain why a pitcher underperforms or how mechanics impact output. In the future, teams that fuse biomechanics with AI will gain a competitive edge — optimizing player development, tailoring strength and rehab programs, and scouting talent based on movement, not just results.